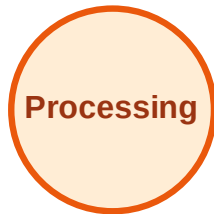


BFS Traversal

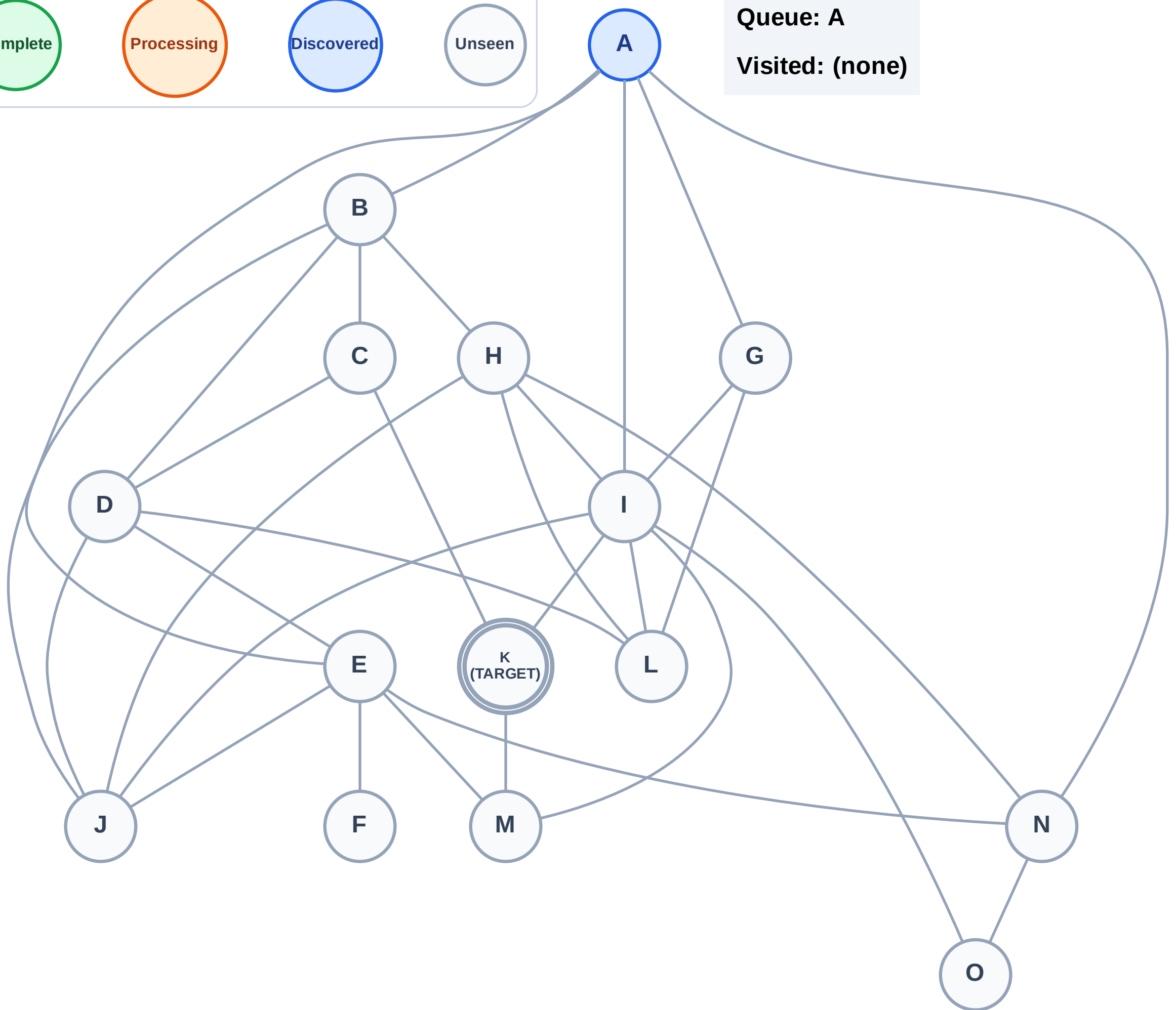
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



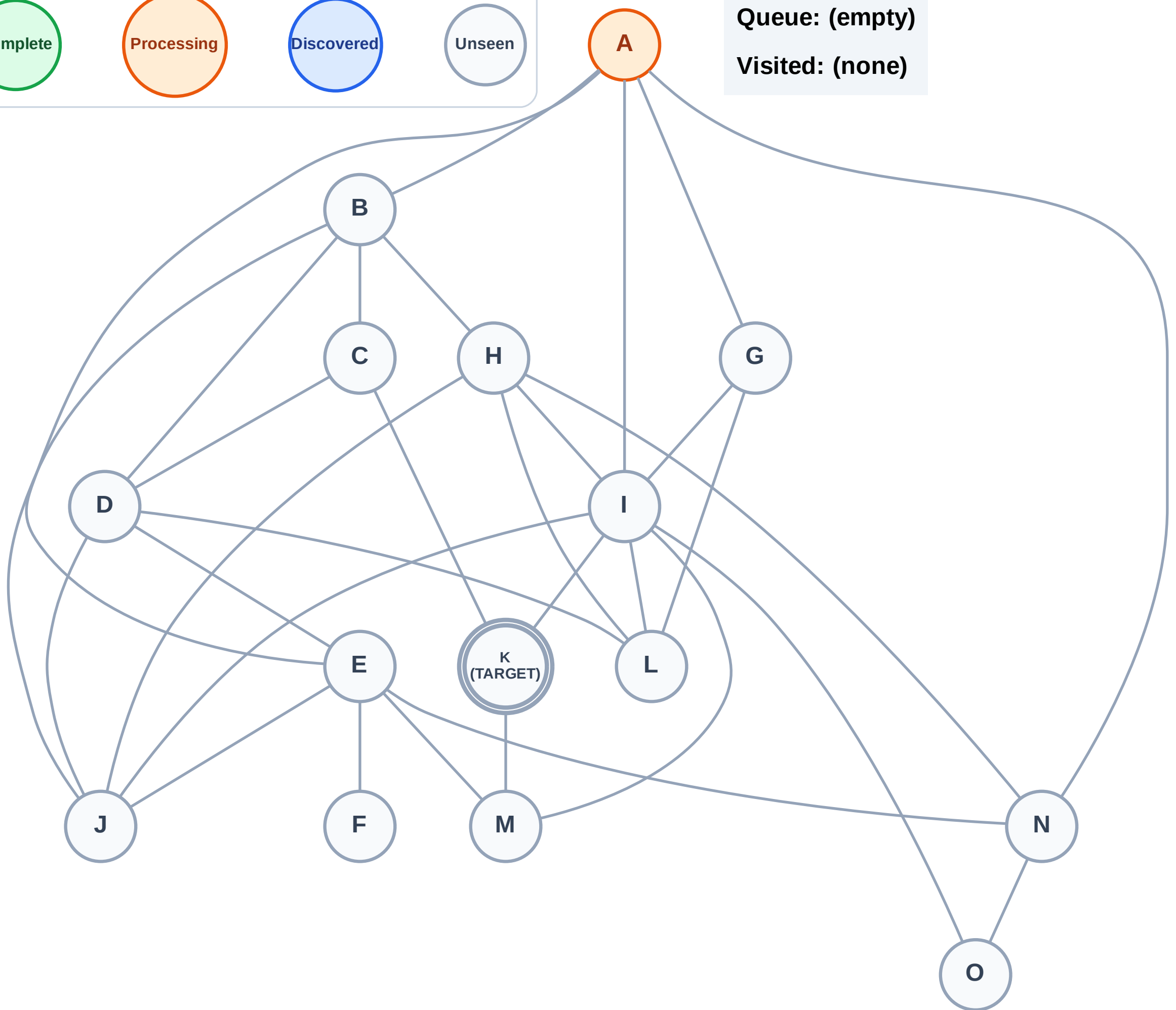
Discovered



Unseen

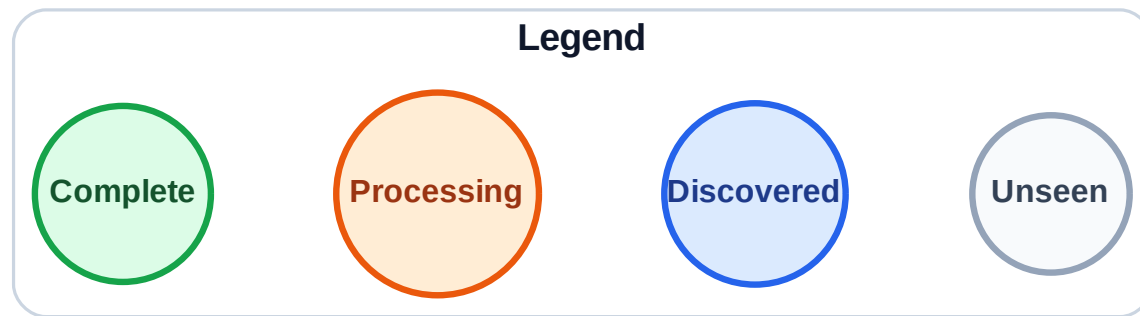
Queue: (empty)

Visited: (none)

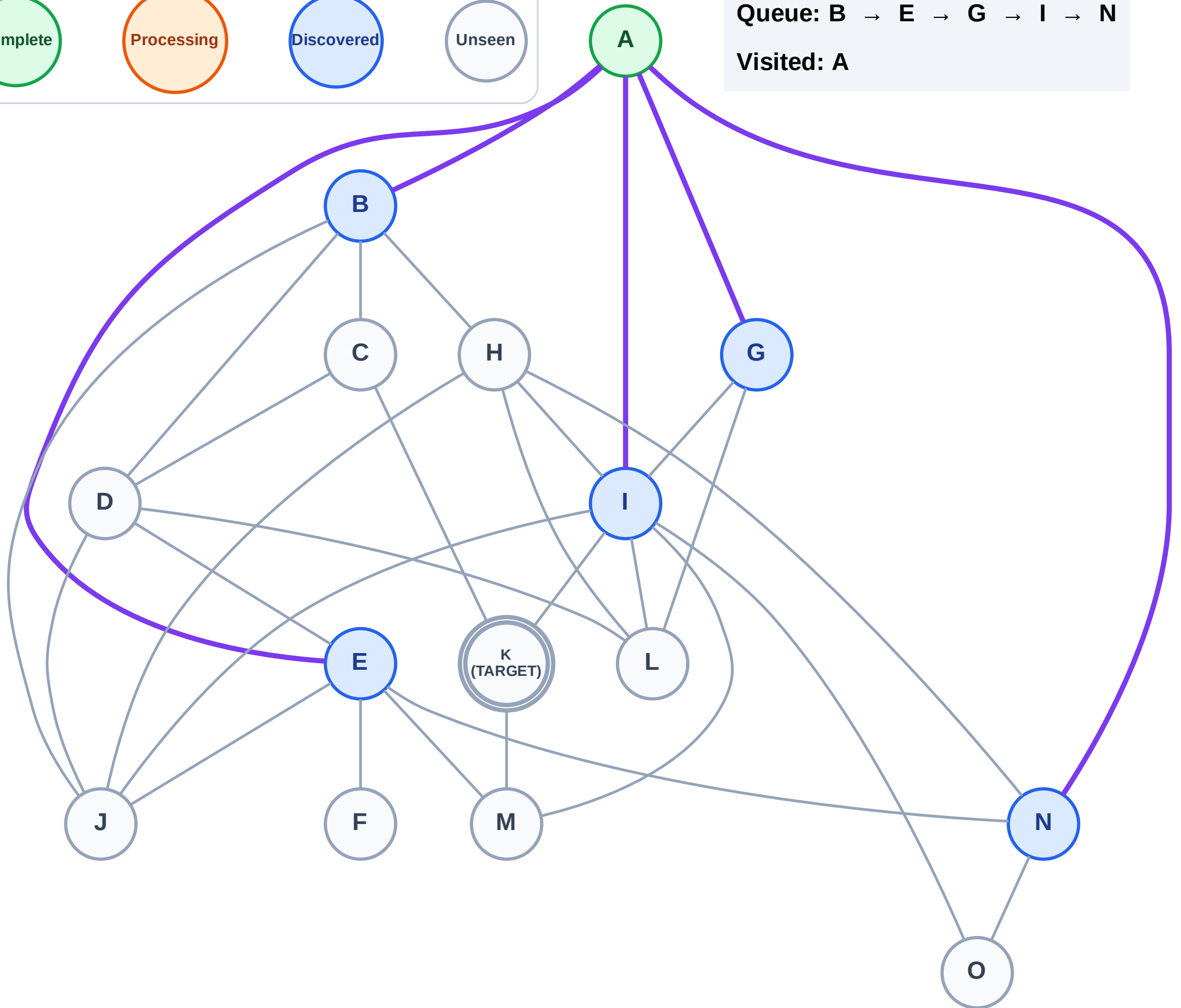


BFS Traversal

Step 3: Discover B, E, G, I, N from A



Queue: B → E → G → I → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

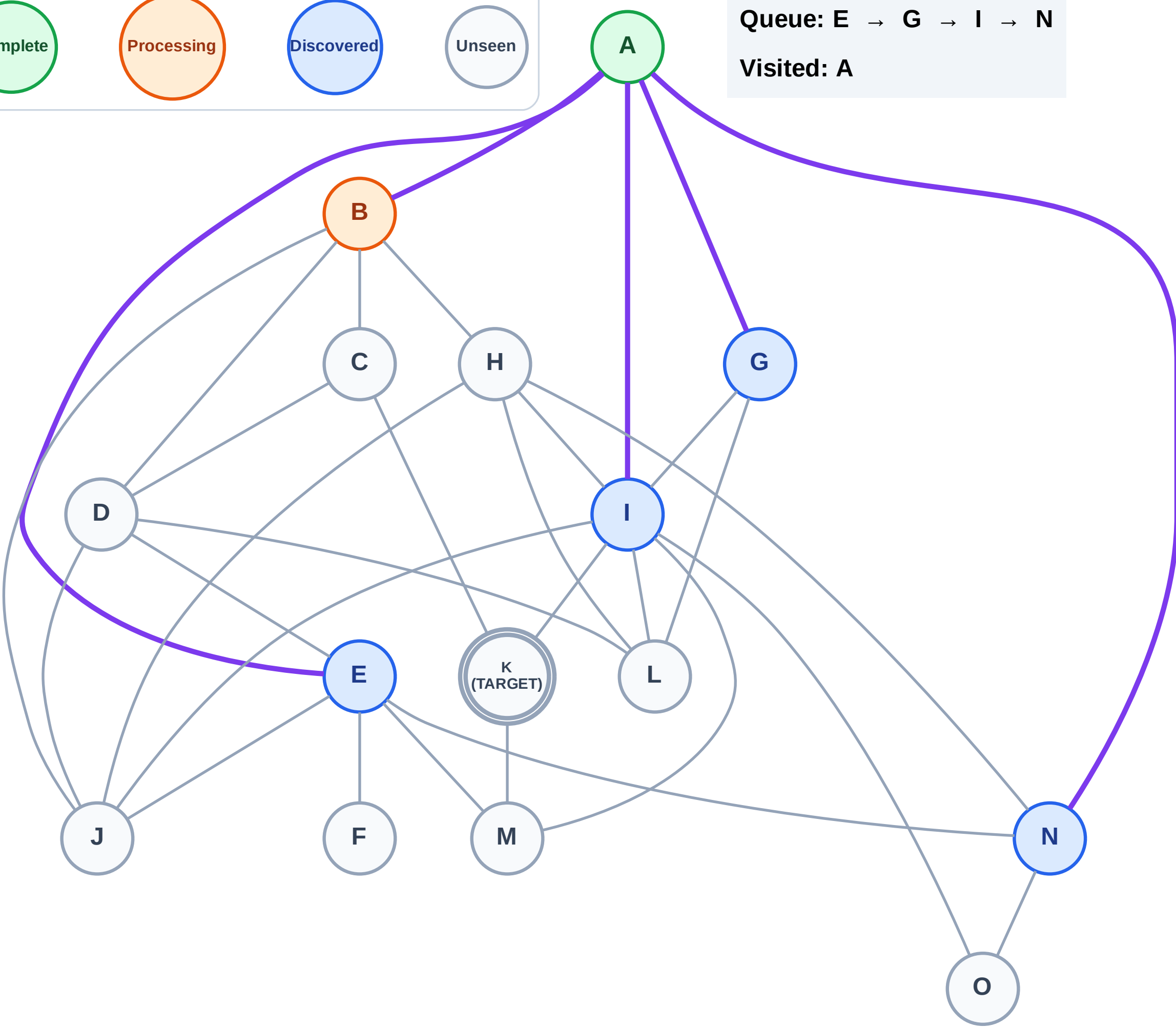
Processing

Discovered

Unseen

Queue: E → G → I → N

Visited: A



BFS Traversal

Step 5: Discover C, D, H, J from B

Legend

Complete

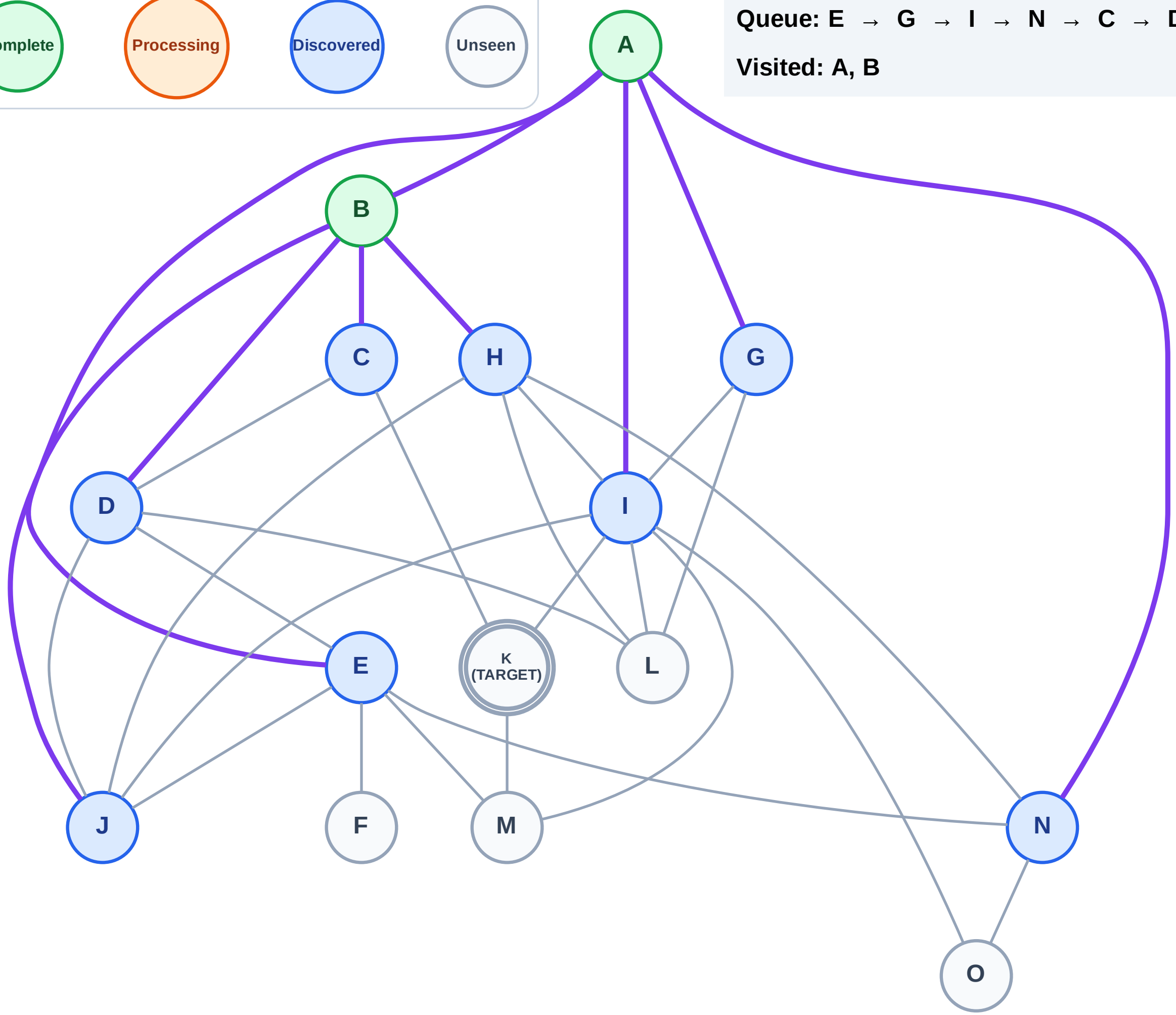
Processing

Discovered

Unseen

Queue: E → G → I → N → C → D → H → J

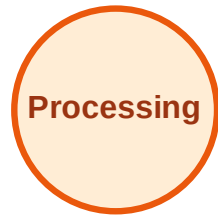
Visited: A, B



BFS Traversal

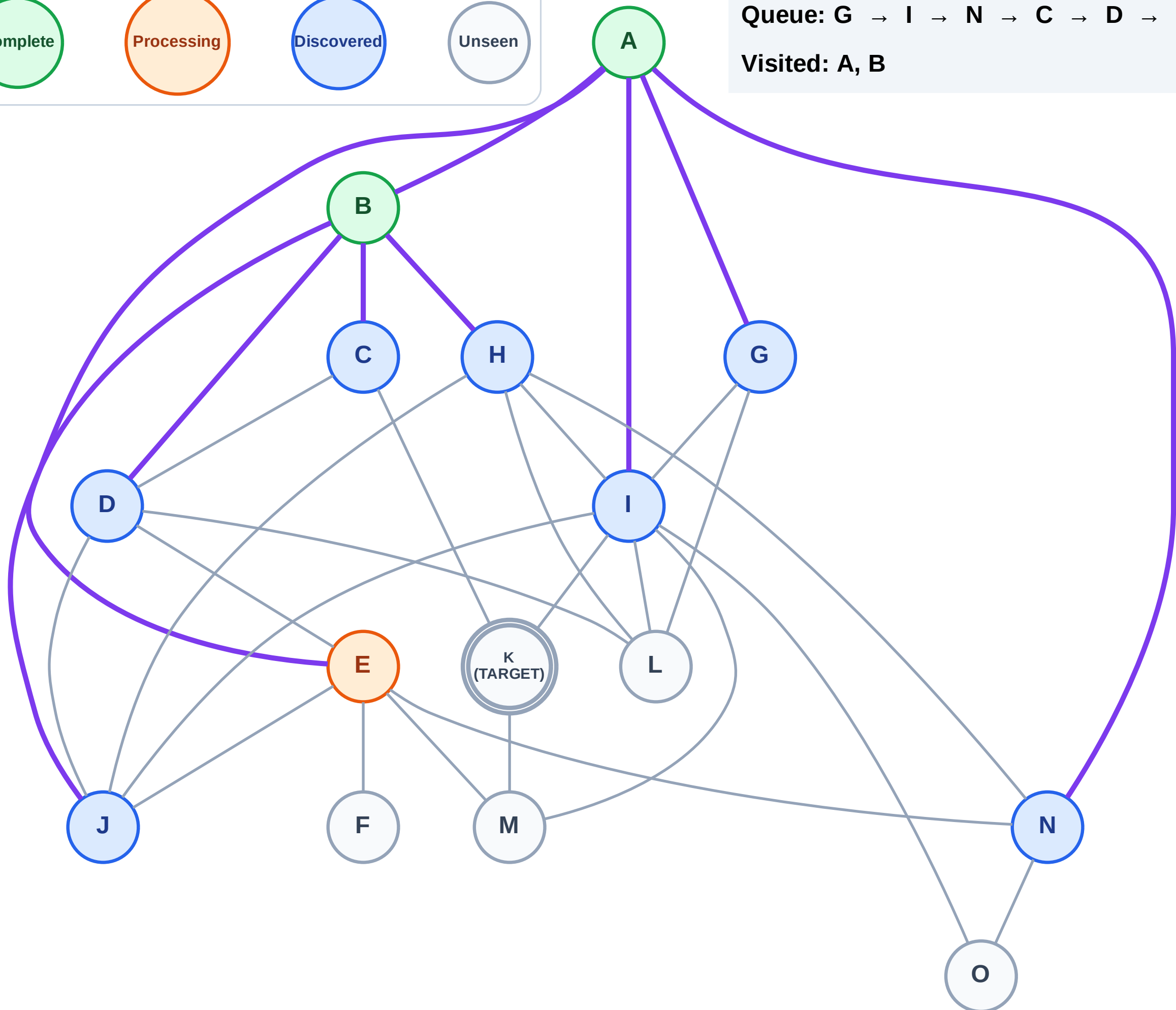
Step 6: Process node E

Legend



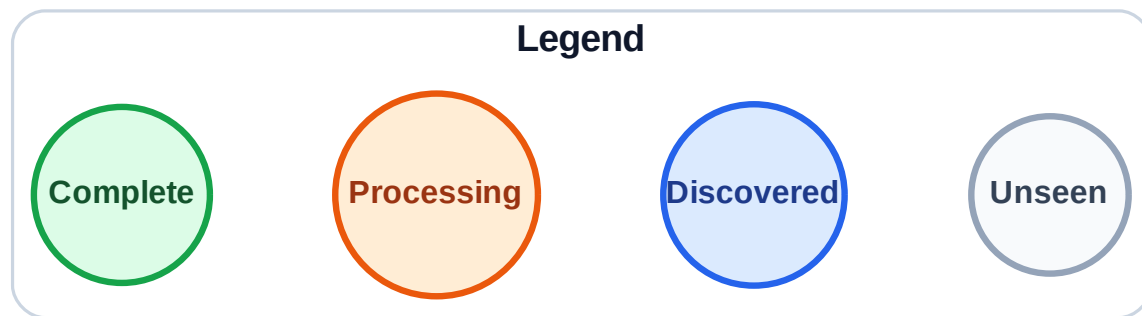
Queue: G → I → N → C → D → H → J

Visited: A, B



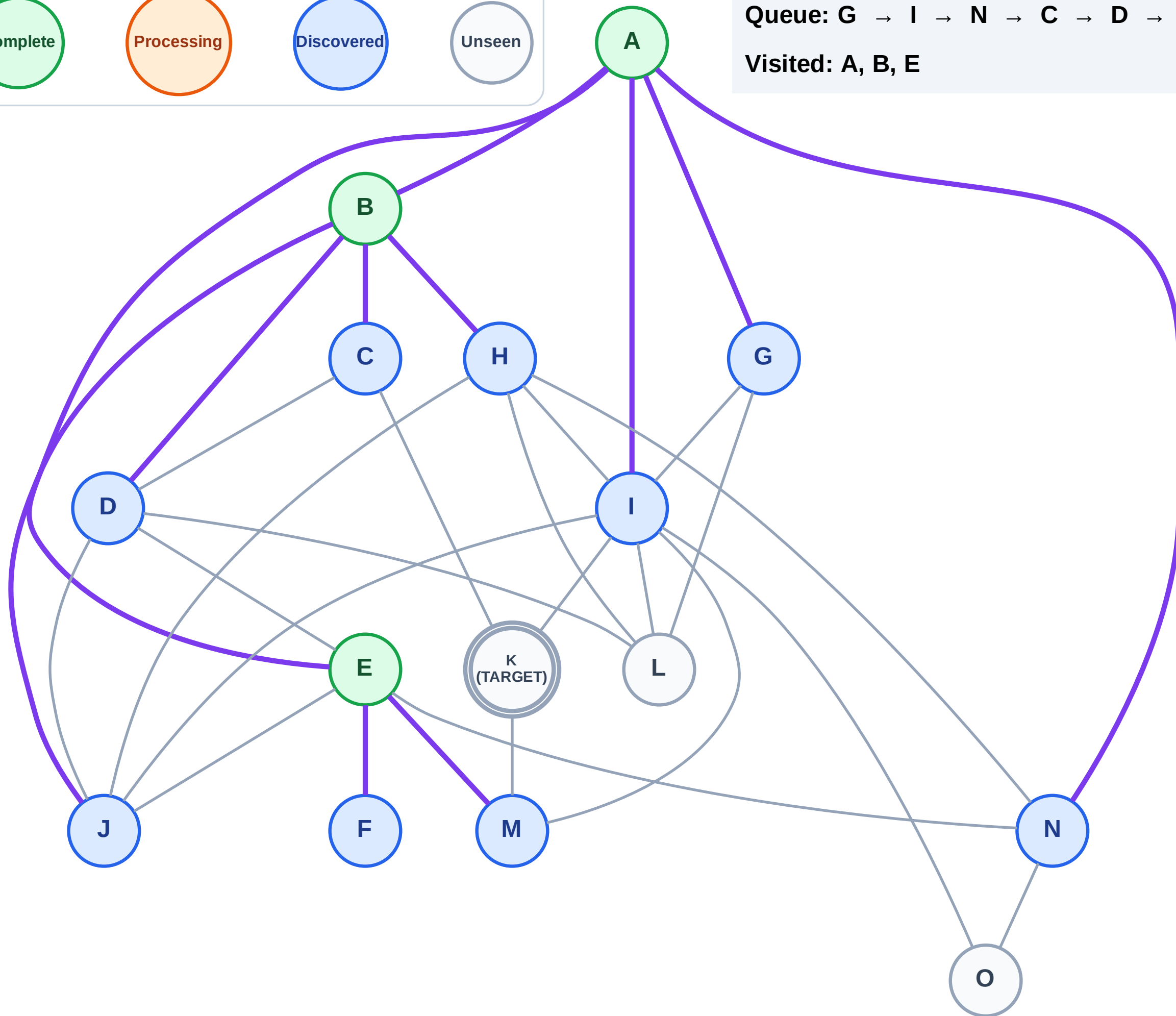
BFS Traversal

Step 7: Discover F, M from E



Queue: G → I → N → C → D → H → J → F → M

Visited: A, B, E



BFS Traversal

Step 8: Process node G

Legend

Complete

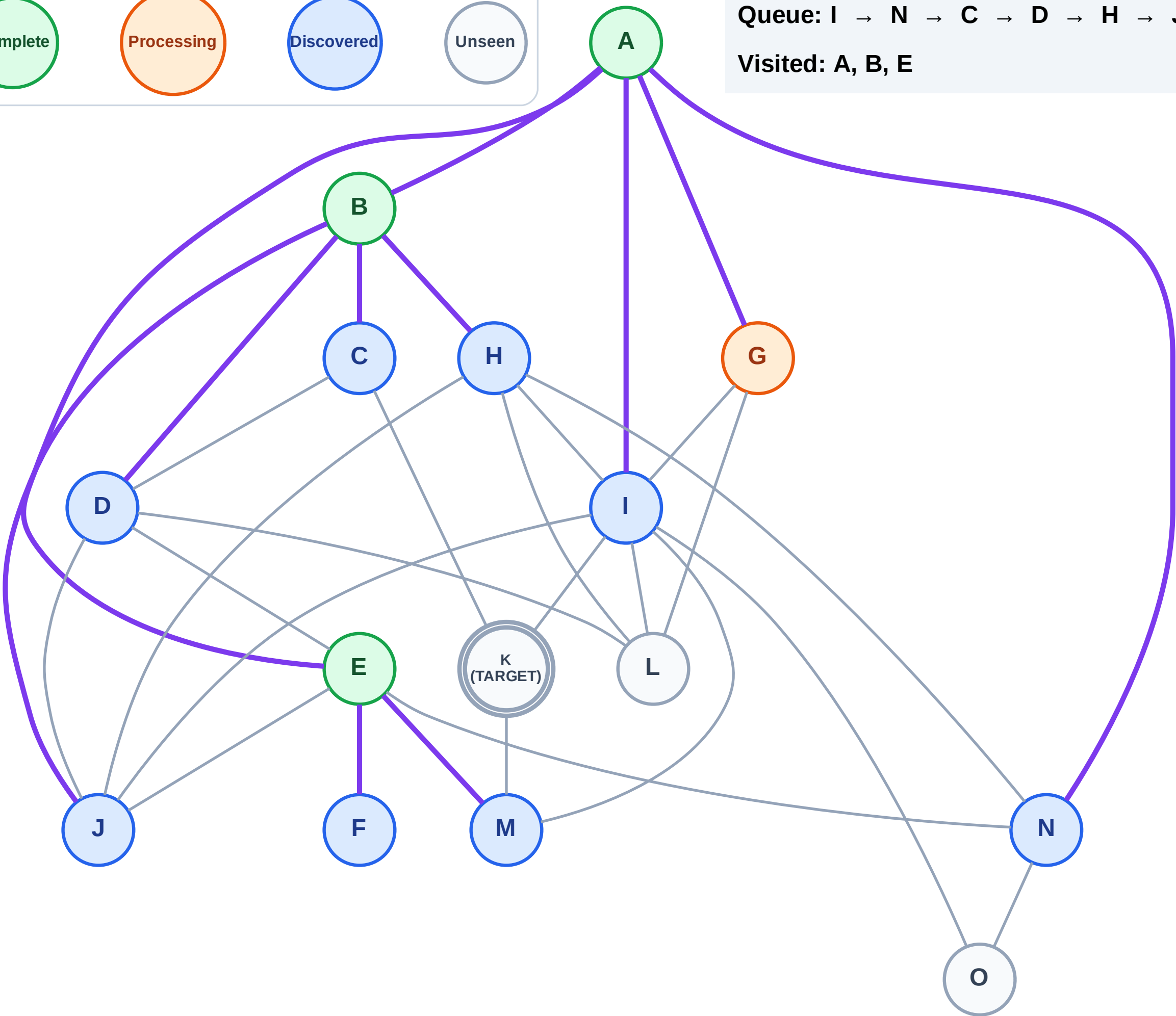
Processing

Discovered

Unseen

Queue: I → N → C → D → H → J → F → M

Visited: A, B, E



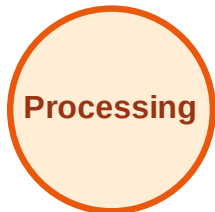
BFS Traversal

Step 9: Discover L from G

Legend



Complete



Processing



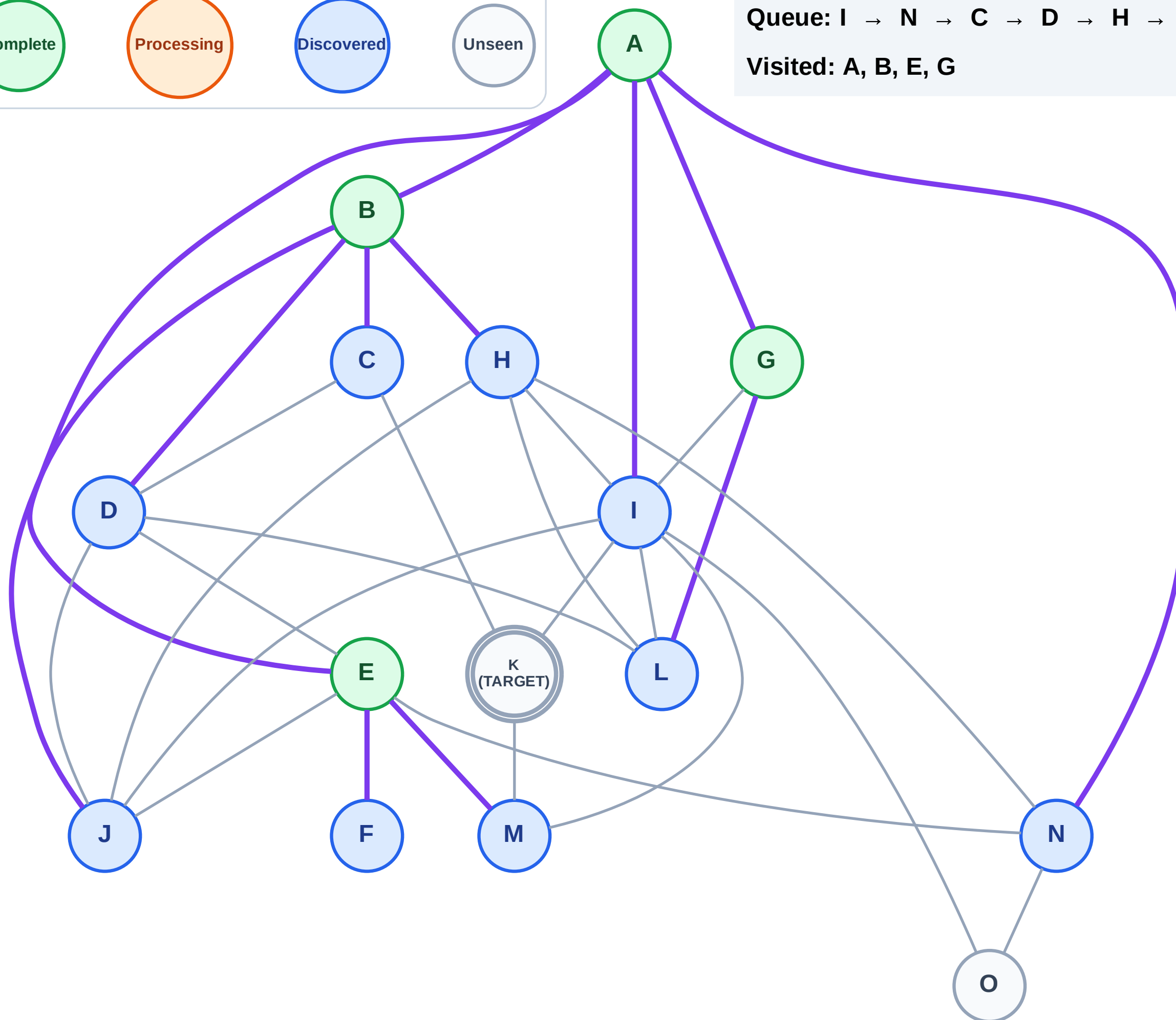
Discovered



Unseen

Queue: I → N → C → D → H → J → F → M → L

Visited: A, B, E, G



BFS Traversal

Step 10: Process node I

Legend

Complete

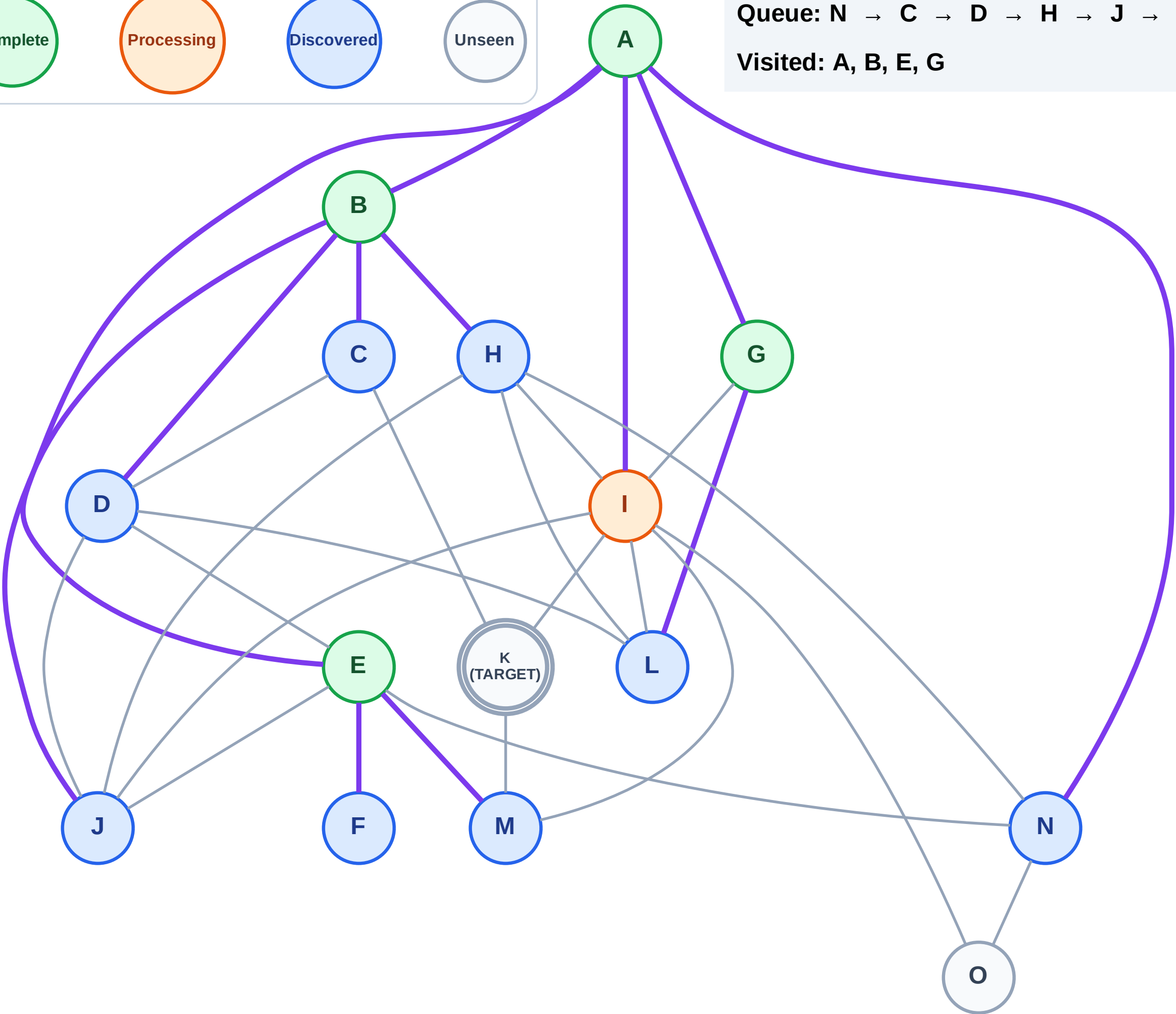
Processing

Discovered

Unseen

Queue: N → C → D → H → J → F → M → L

Visited: A, B, E, G



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend

Complete

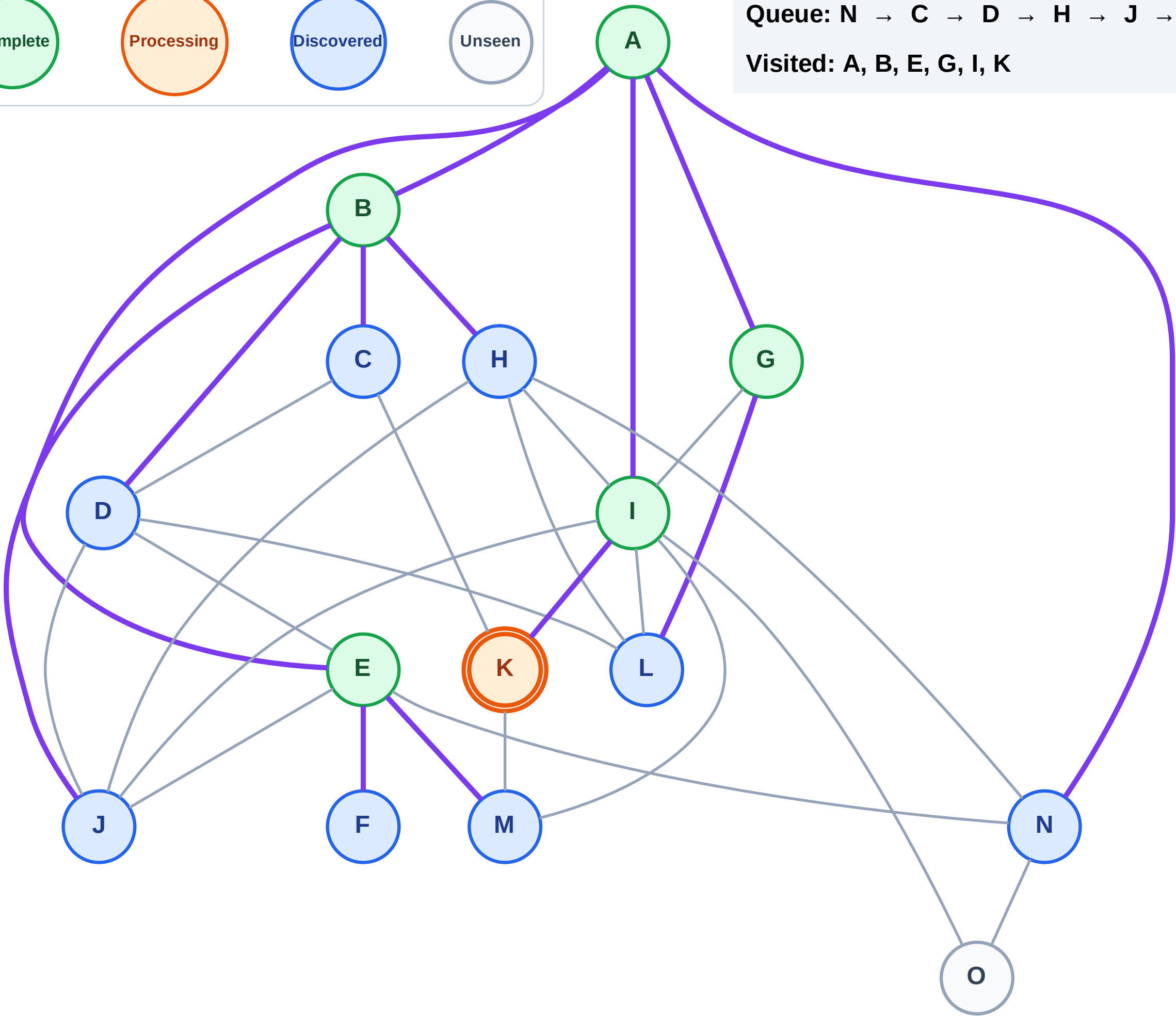
Processing

Discovered

Unseen

Queue: N → C → D → H → J → F → M → L

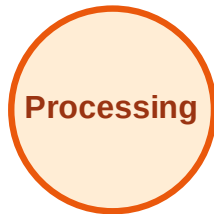
Visited: A, B, E, G, I, K



DFS Traversal

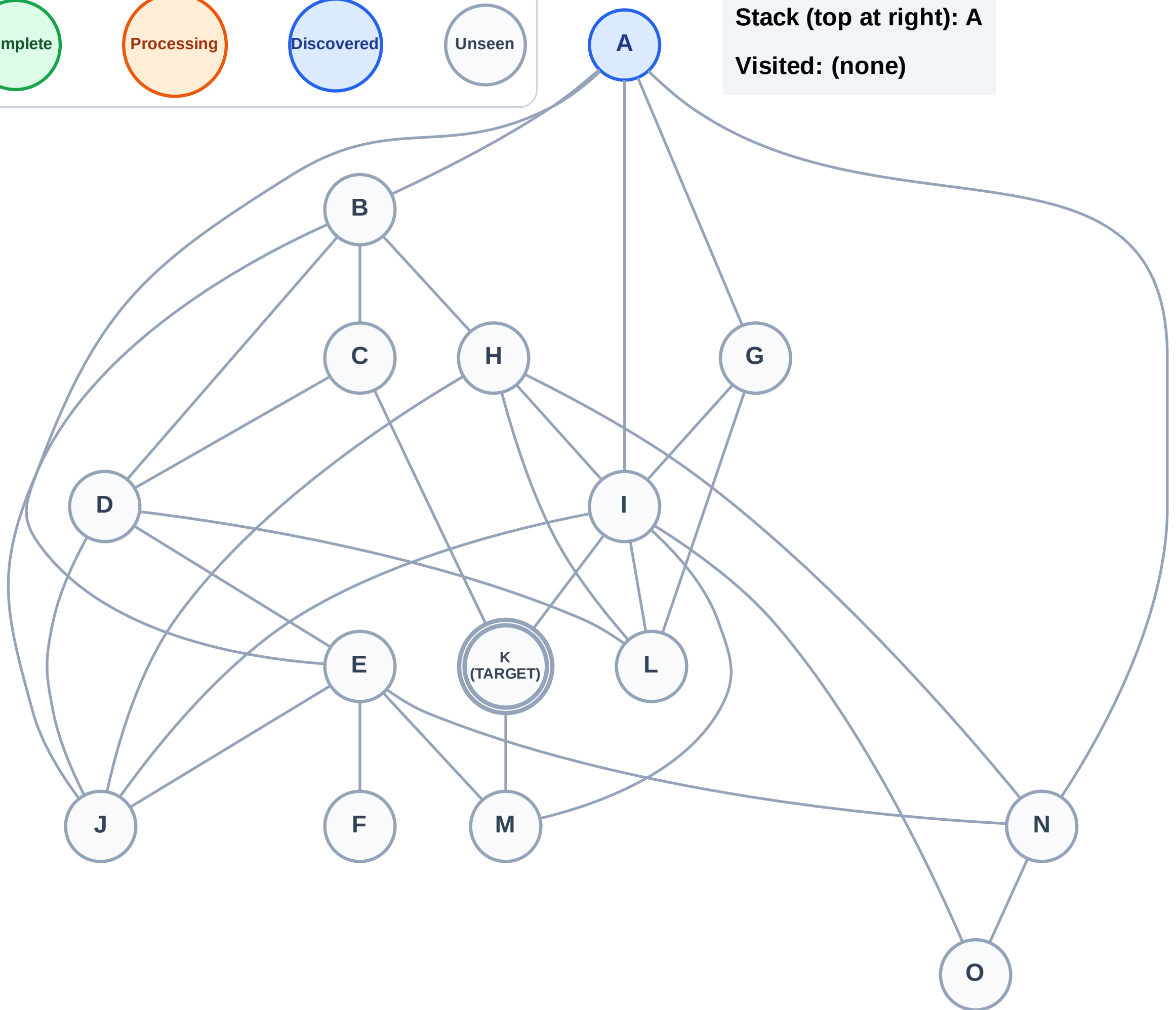
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

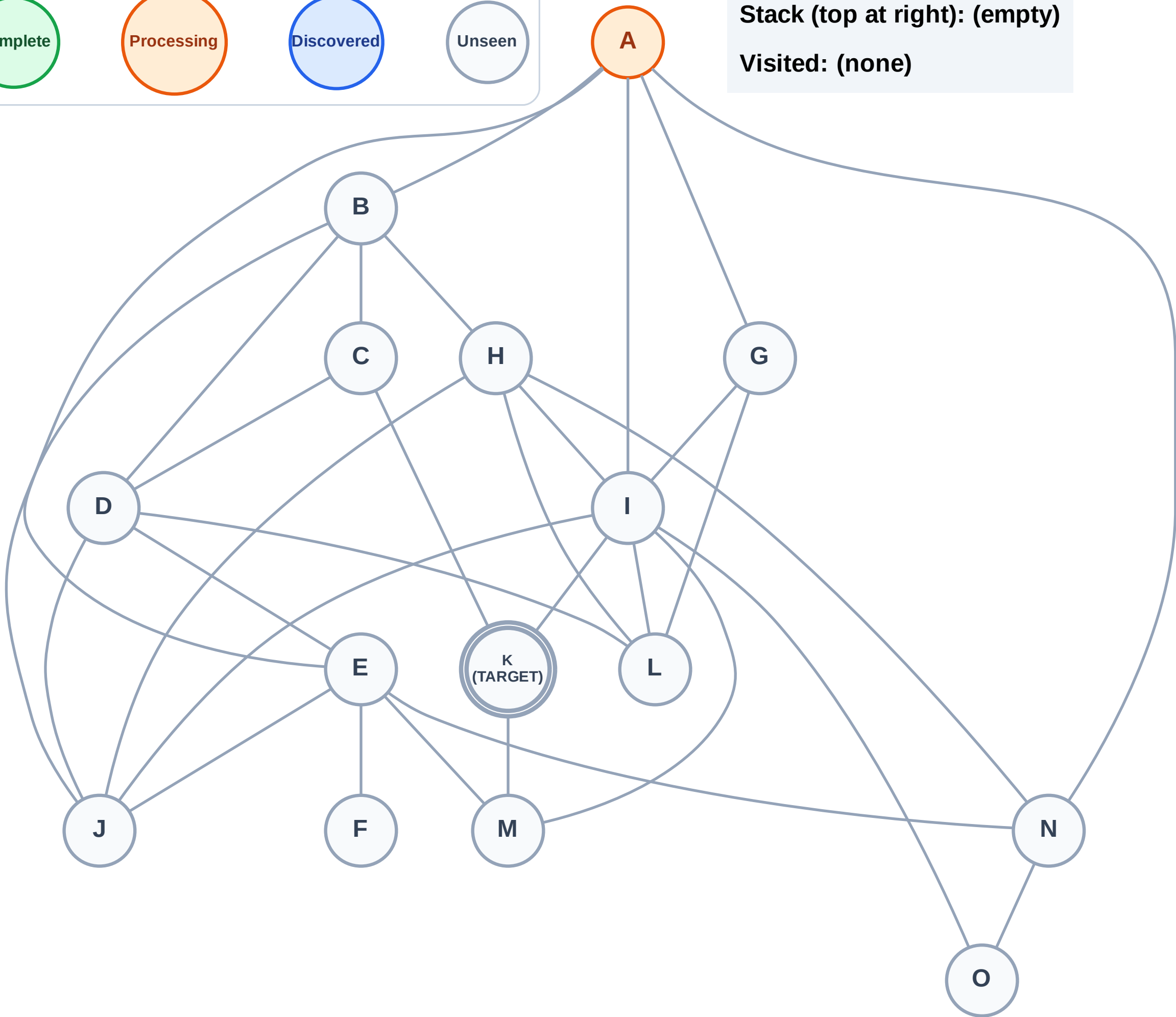
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover N, I, G, E, B from A

Legend

Complete

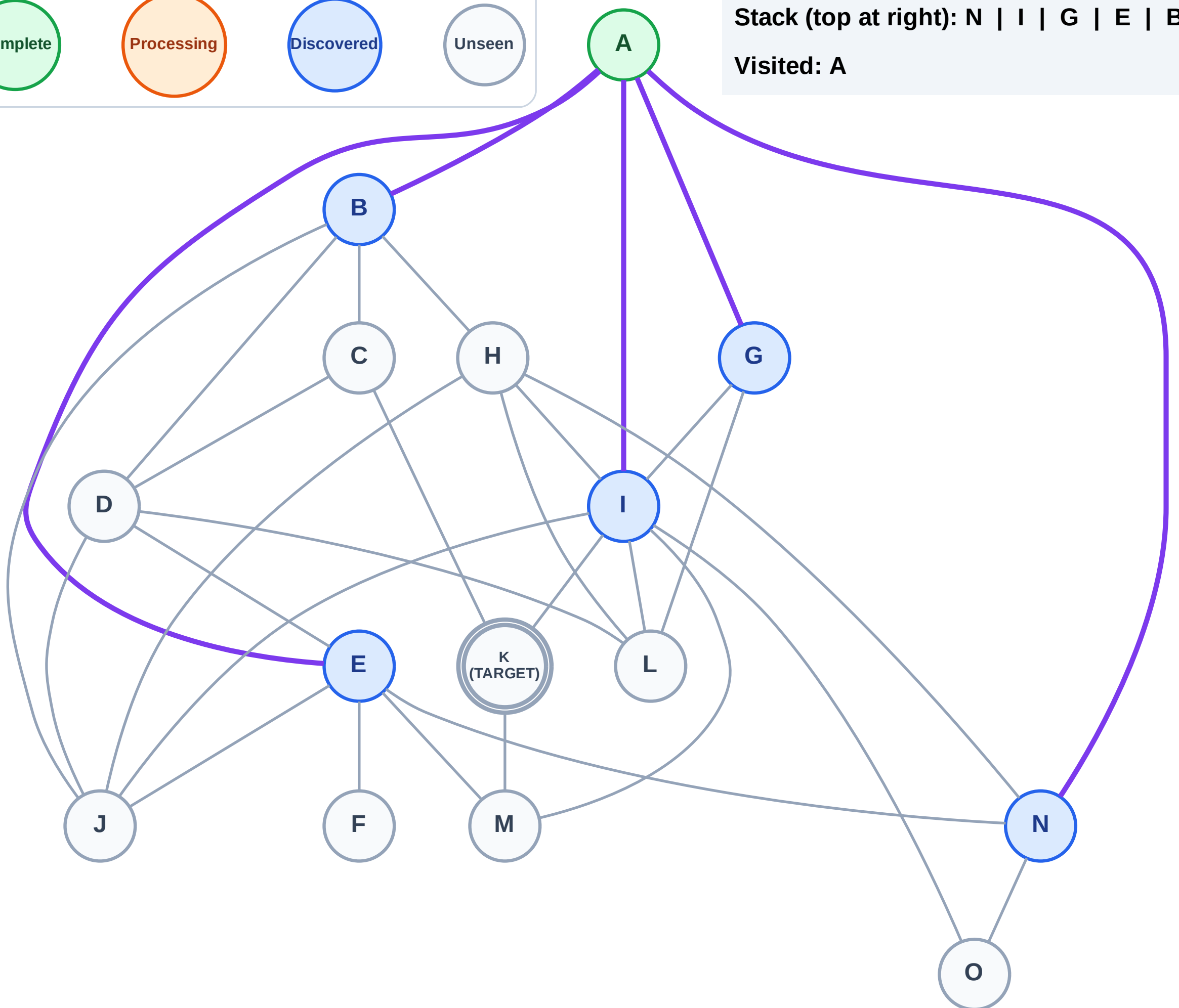
Processing

Discovered

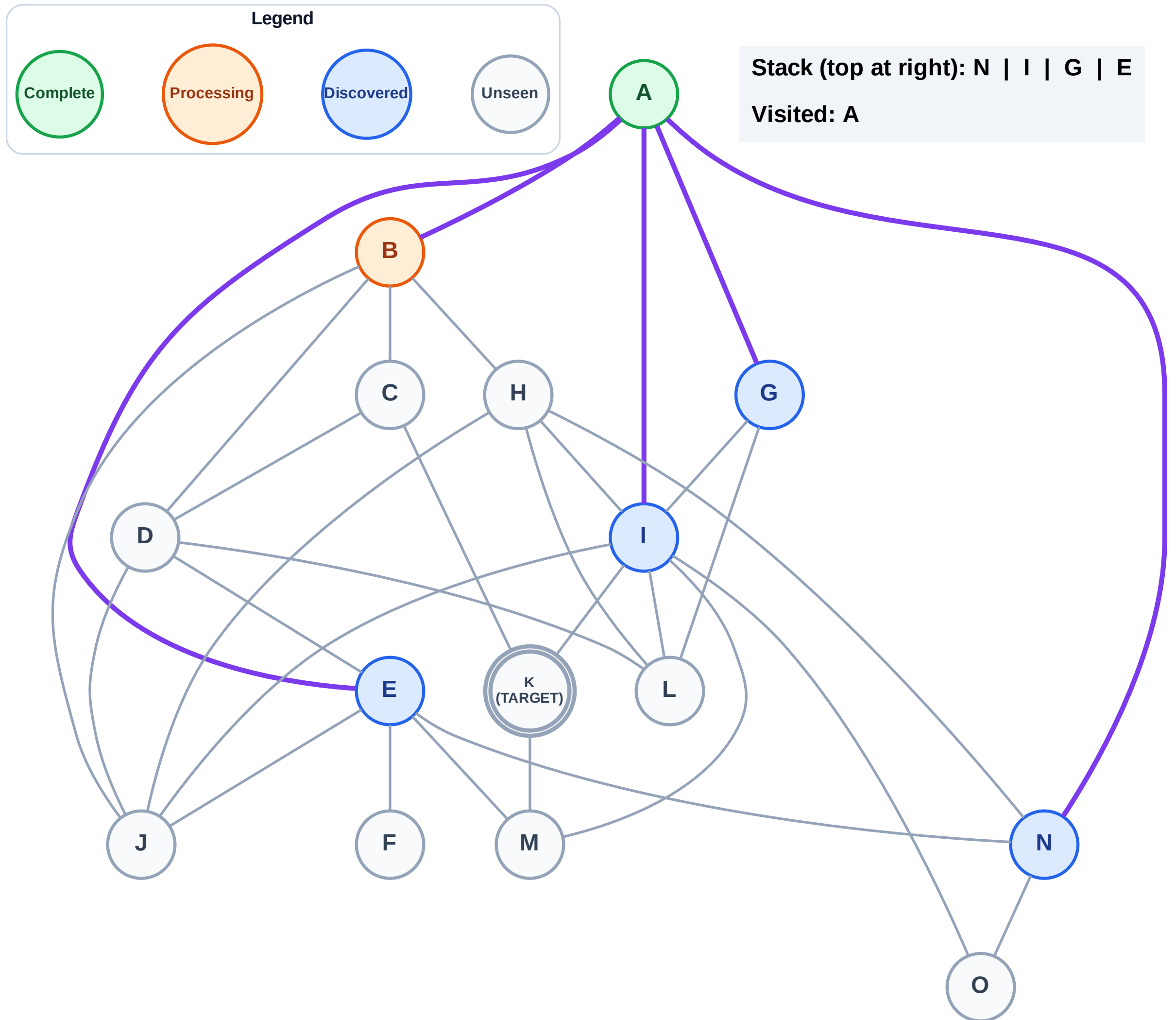
Unseen

Stack (top at right): N | I | G | E | B

Visited: A



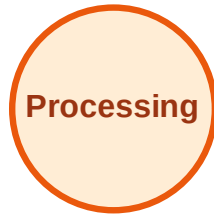
Step 4: Process node B



DFS Traversal

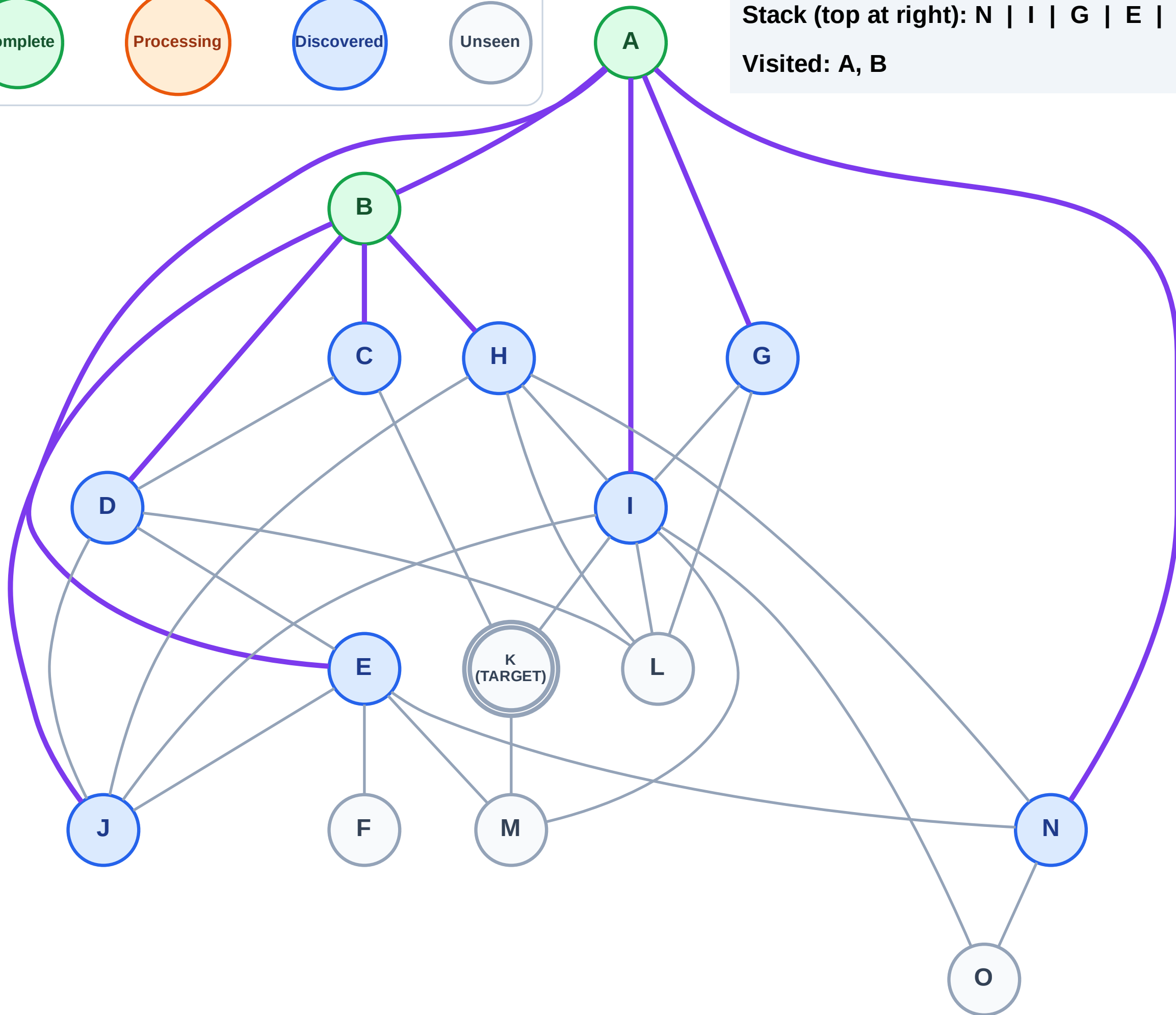
Step 5: Discover J, H, D, C from B

Legend



Stack (top at right): N | I | G | E | J | H | D | C

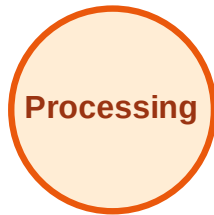
Visited: A, B



DFS Traversal

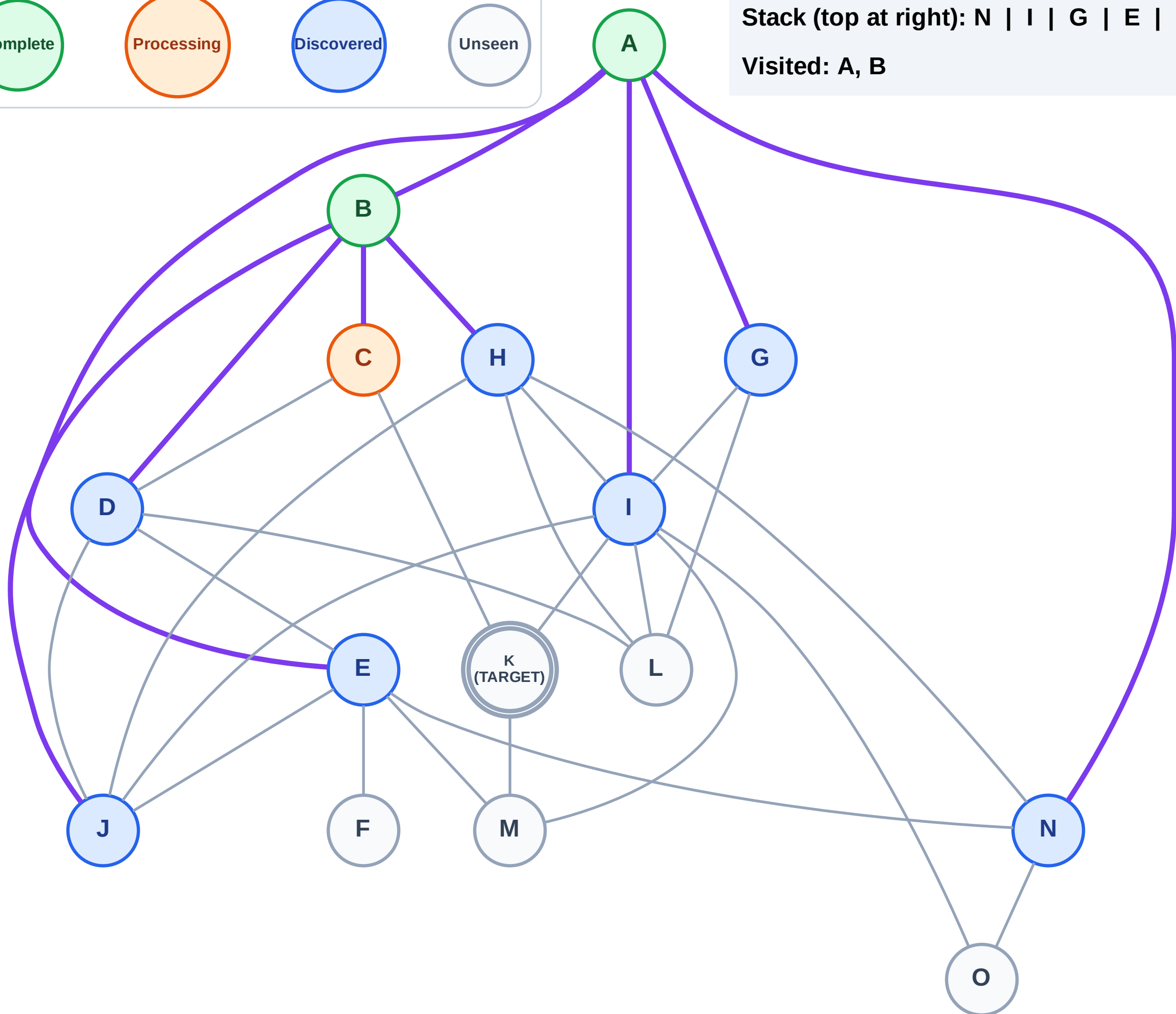
Step 6: Process node C

Legend



Stack (top at right): N | I | G | E | J | H | D

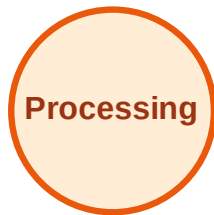
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Stack (top at right): N | I | G | E | J | H | D

Visited: A, B, C, K

